

David Elijah Umaru

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SUMMARY

Computer Engineering senior (B.S., May 2026, GPA 3.94) applying to graduate study in Computer Engineering, with two years leading electrical systems for Formula SAE Ram Racing's electric and combustion vehicles. Experience spans 400V tractive-system integration, CAN bus architecture, custom safety PCBs, and FSAE-compliant wiring harnesses, backed by a Tesla battery-program sponsorship.

EDUCATION

Colorado State University — Fort Collins, CO

May 2026

B.S. Computer Engineering, Minor in Computer Science — Concentration: Embedded Systems

GPA: 3.94/4.0 (Dean's List, 6 semesters)

EXPERIENCE

HPC Developer Intern

June 2025 – Present

Data Science Research Institute, Colorado State University

- Supported high-availability Linux/HPC systems and assisted with commissioning during a data-center rebuild, including configuration checks and operational testing.
- Authored internal technical documentation (GitHub, Read the Docs, Freshservice), improving support efficiency ~50%.

IT Student Support & A/V Media Specialist

June 2023 – Present

College of Business, Colorado State University

- Resolved IT incidents for 275+ faculty/staff, deployed/imaged 150+ Windows systems, and led Crestron/AV podium renovations, training 50+ end users.

Automation Intern

August 2022 – December 2022

Dept. of Microbiology, Immunology, and Pathology, Colorado State University

- Developed Python automation scripts using the Opentrons API, increasing laboratory throughput ~50% while reducing manual error.

PROJECTS

Formula SAE Ram Racing (RR26EV) — Low Voltage Electrical Lead

Sept 2024 – Present

KiCad, CAN bus/DBC, MATLAB/Simulink, Deutsch/Amphenol connectors

- Designed the complete 12V low-voltage architecture (6 independently-fused channels) and a 500kbps CAN bus network (7 nodes, custom DBC file) integrating the motor controller, BMS, IMD, BSPD, and VCU for an electric FSAE vehicle; passed FSAE 2026 compliance review at 95% on first submission.
- Designed 3 custom PCBs (power distribution, signal conditioning, fault-latch) in KiCad with IPC-2221-compliant trace widths, cutting hardware cost ~\$1,300 vs. commercial alternatives and achieving <5ms shutdown response time.
- Engineered and fabricated 5 wiring harness assemblies (85m of wire, 120+ crimped connections) to FSAE T.9.2.2 gauge/connector specs; passed vibration testing (5G, 10–500Hz) and a 10-hour endurance test with zero CAN errors.
- Developed vehicle control logic (MATLAB/Simulink) interfacing the New Eagle RCM120, EMRAX resolver, and PM150DX-SP motor controller, reducing integration time 20%.
- Coordinated sponsor relationships (secured a New Eagle hardware/software sponsorship) and trained 10+ new electrical team members.

Rubicon — Electrical Lead, FSAE EV Prototype

2024 – 2025

CAN telemetry, motor controller calibration, benchtop validation

- Designed and fabricated the complete low-voltage wiring harness and a CAN-based driver dashboard (battery voltage, motor temp, torque, faults) for Ram Racing's first EV prototype.
- Built a benchtop powertrain test stand (E-stop, precharge circuit, CAN fault-injection framework) validating motor-controller startup sequencing and thermal derating before vehicle integration.
- Calibrated motor-controller current limits and acceleration curves; established LV harness and CAN protocol standards adopted for the 2025-26 competition build.

Intrepid — Wireless Telemetry System, FSAE IC Car

2024

Teensy 4.1, dual CAN bus, 915MHz RF, Node-RED

- Designed a wireless telemetry system (Teensy 4.1, dual CAN buses, 915MHz RF link, Raspberry Pi/Node-RED dashboard) streaming Haltech ECU data to the pit at FSAE Michigan 2024 with <200ms latency and zero data loss, cutting RF bandwidth usage 60% via packed-binary encoding and priority-based scheduling.

TECHNICAL SKILLS

Automotive & Embedded: CAN bus, UART, SPI, PCB design (KiCad), automotive wiring harnesses, Teensy 4.1, ESP32, Raspberry Pi
Languages & Tools: C, C++, Python, MATLAB/Simulink, VHDL, Git/GitHub, Quartus Prime, Onshape, Linux

RELEVANT COURSEWORK

Electrical Circuit Analysis, Digital Circuit Logic, Digital System Design, Signals and Systems, Embedded Systems and Machine Learning, VLSI Design

LEADERSHIP & ACTIVITIES

- Electrical Low Voltage Lead, CSU Formula SAE Electric (2025–2026) — led a 7+ engineer electrical team through the program's first competition EV build.
- Electrical Systems Lead, CSU FSAE EV Electric (2024–2025) — led harness design and motor calibration; cut a 6-week timeline to 2 weeks before Vehicle Reveal.
- Safety Officer, RamBots CSU Senior Design (2025–2026) — owned safety protocols and documentation for a 13-person team.
- Member, National Society of Black Engineers (2023–2026) and ColorStack.

AWARDS & HONORS

Tesla Battery Sponsorship Recipient (2025) · New Eagle Sponsorship (secured) · Fry Family Electrical Engineering Scholarship · Carl Wilsen Scholarship in ECE · Dean's List (6 semesters)